

# Marco Antonio Bacchetta Villalba

Entry-Level Backend Developer | TypeScript | Node.js | API Architecture

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## PROFESSIONAL SUMMARY

Entry-Level Backend Developer transitioning from a legal career, bringing analytical precision, compliance awareness, and structured problem-solving into software engineering. Focused on scalable, type-safe API architectures, clean server-side logic, and maintainable backend systems using TypeScript, Node.js, Express, SQL, and Firebase. Strong foundation in algorithmic problem-solving, data structures, and optimized logic design. Upper-intermediate English (B2), comfortable in international and cross-functional environments.

## CORE TECHNICAL SKILLS

|                      |                                                                                                  |
|----------------------|--------------------------------------------------------------------------------------------------|
| Languages            | TypeScript, JavaScript ES6+, SQL                                                                 |
| Backend Development  | Node.js, Express, API Architecture, CRUD Operations, Firebase Integration, CLI Development       |
| Computer Science     | Data Structures & Algorithms, Two-Pointer Technique, Array Manipulation, Complexity Optimization |
| Tools & Environments | Git, GitHub, tsx, Node.js fs Module, JSON Persistence, Local AI Tools/Hardware Workflows         |

## TECHNICAL PROJECTS

### Baby Shower Gift Registry Application - Backend / Full-Stack Developer, April 2026

- Engineered a real-time gift registry system by implementing full CRUD functionality for gift creation, updates, reservations, and deletion.
- Integrated Firebase for persistent data storage and authentication, enabling reliable backend state management.
- Improved data consistency across concurrent user requests by applying state validation and controlled update logic.

### Task Tracker CLI - TypeScript / Node.js Developer

- Architected a command-line task manager with full CRUD operations, status transitions, and filtered task views using TypeScript and Node.js.
- Implemented persistent local storage with the native fs module, JSON.parse(), and formatted JSON.stringify(data, null, 2) writes to tasks.json.
- Strengthened runtime safety by validating inputs, handling empty datasets, using strict typing, and applying array methods such as filter(), findIndex(), and at(-1).

### Algorithmic Optimization & API Challenges - Independent Developer

- Optimized complex array and string-processing challenges by applying advanced algorithmic patterns, including the two-pointer technique.
- Solved problems such as Trapping Rain Water with linear-time logic, targeting O(N) efficiency and reduced auxiliary space.
- Translated abstract constraints into clean, testable, and maintainable server-side logic.

## PROFESSIONAL EXPERIENCE

### Legal Professional / Attorney - Asunción, Paraguay | Previous Experience

- Analyzed complex regulatory frameworks and drafted structured legal documentation using logical frameworks comparable to software specifications.
- Managed compliance and dispute resolution protocols, ensuring data integrity, procedural accuracy, and adherence to operational rules.
- Collaborated with cross-functional stakeholders and international bodies, supporting planning, logistics, communication, and workflow optimization.

## EDUCATION & CERTIFICATIONS

Bachelor's Degree in Law - Facultad de Derecho y Ciencias Sociales, Universidad Nacional de Asunción - UNA

Continuous Specialized Training - Backend Software Architecture, Node.js/TypeScript Ecosystems, Advanced Data Structures & Algorithms

## COMMUNITY & LEADERSHIP

- Active Member, Google Developer Group - GDG Asunción:** Engaged in tech networking, backend architecture discussions, and developer community events.
- Board Member, Sociedad Suiza del Paraguay:** Supports administration, compliance, governance, and strategic planning.